Muscles & Neurovascular Structures around the Pectoral Girdle and Arm



Ngozi Onuegbu PhD

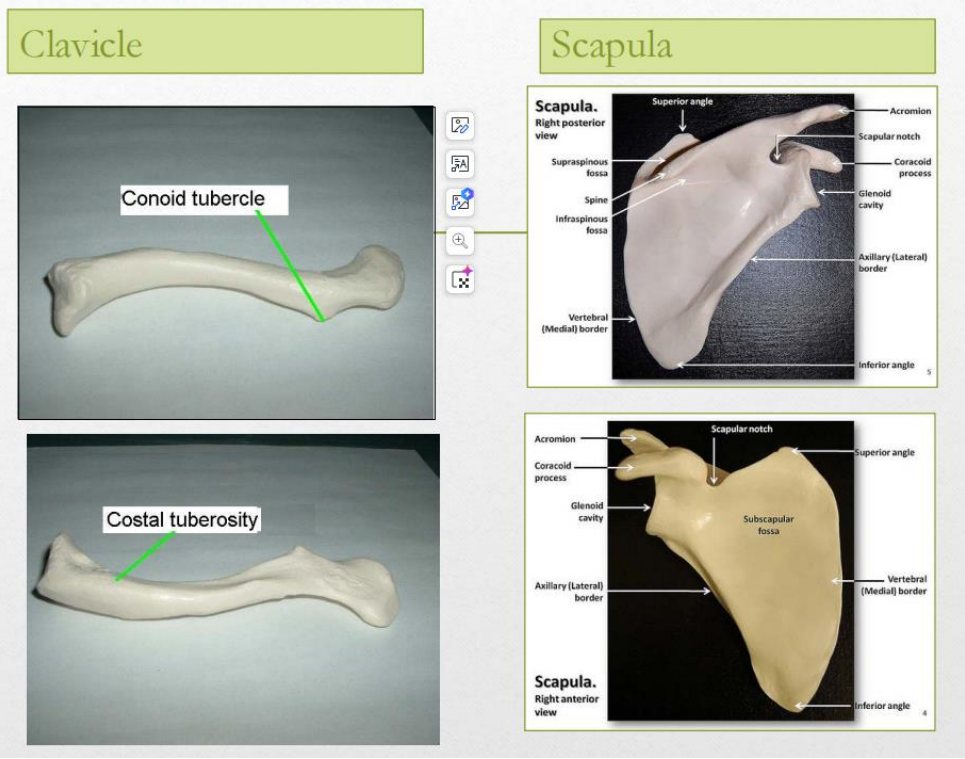
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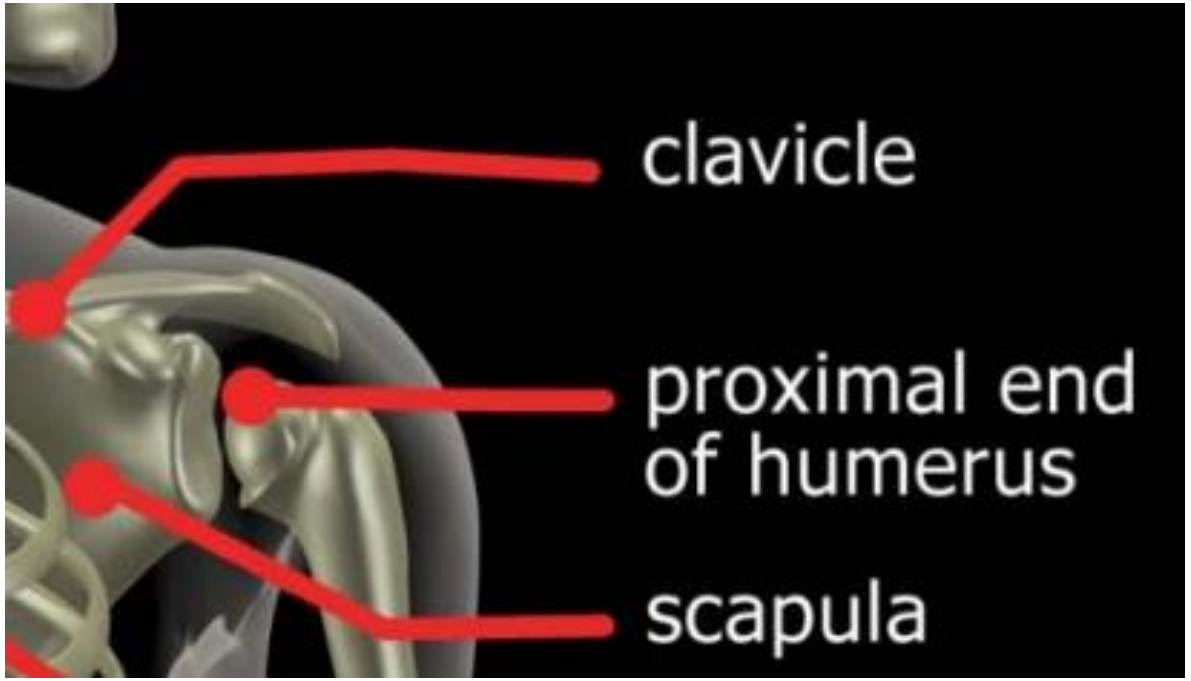
Learning Objectives

- By the end of this session, students should be able to:

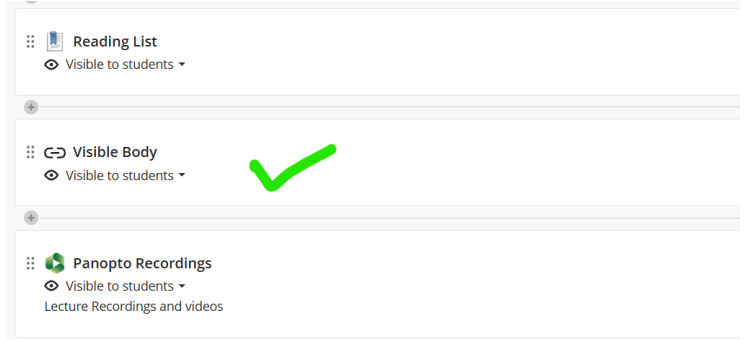
1. Identify the major muscles of the pectoral girdle and arm, including their origins, insertions, and functions.
2. Describe the innervation and vascular supply of these muscles, correlating them with their respective nerves and arteries.
3. Apply knowledge of muscle function and neurovascular supply to clinical assessments, injury evaluation, and rehabilitation strategies.

Pectoral Girdle: Shoulder Girdle



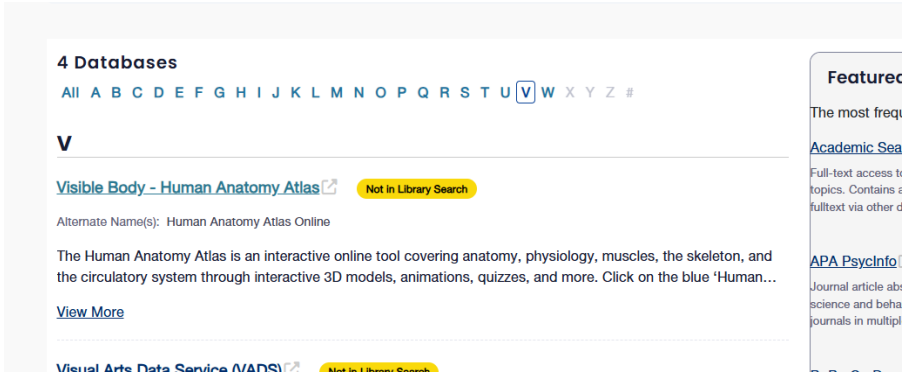


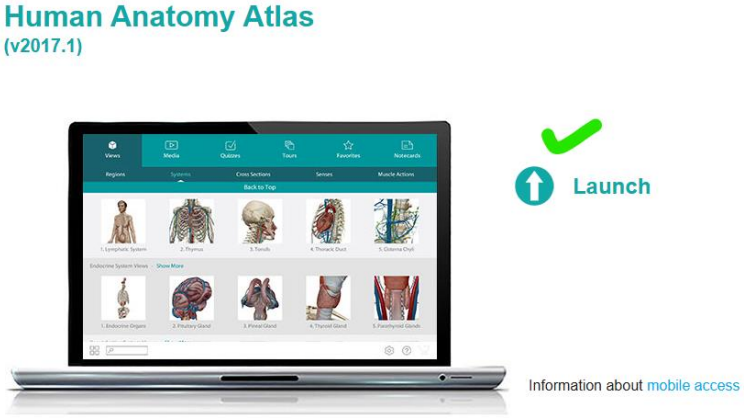
Practical Instructions



- In your module, click on the Visible Body Link

- Launch the *Visible Body – Human Anatomy Atlas*

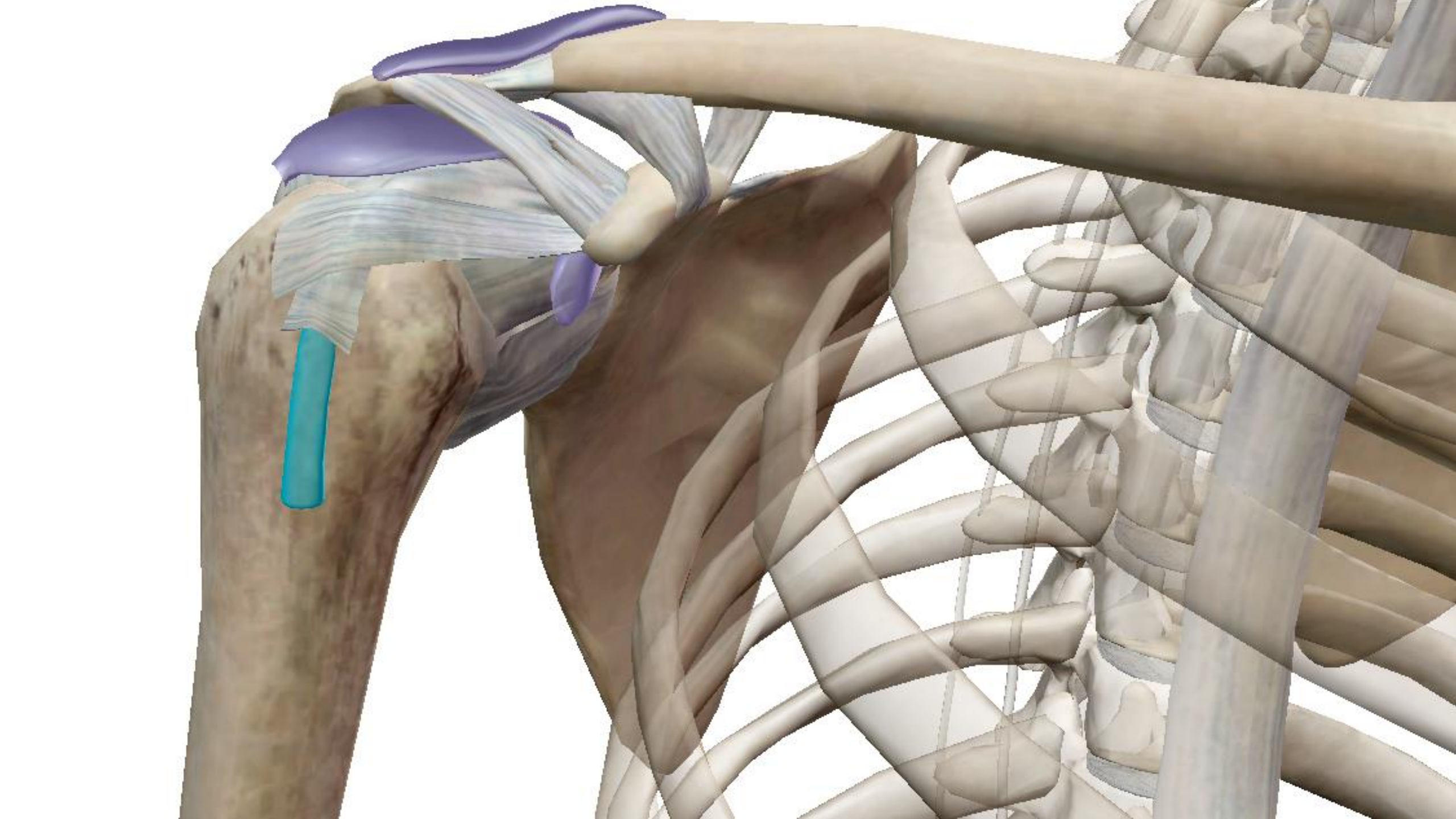




Skeletal System Views ➔ Show More ➔ Shoulder Girdle

In your groups:

- Identify the bones of the Shoulder girdle
- Make use of the select tool and interact with the visible body system (3-D interaction)
- Identify the ligaments that hold the bones together
- Compare the images and match them to what you learned in Anatomy of the Shoulder Joint earlier today.
- Explore the relationship of the pectoral girdle bones with the thoracic cage.

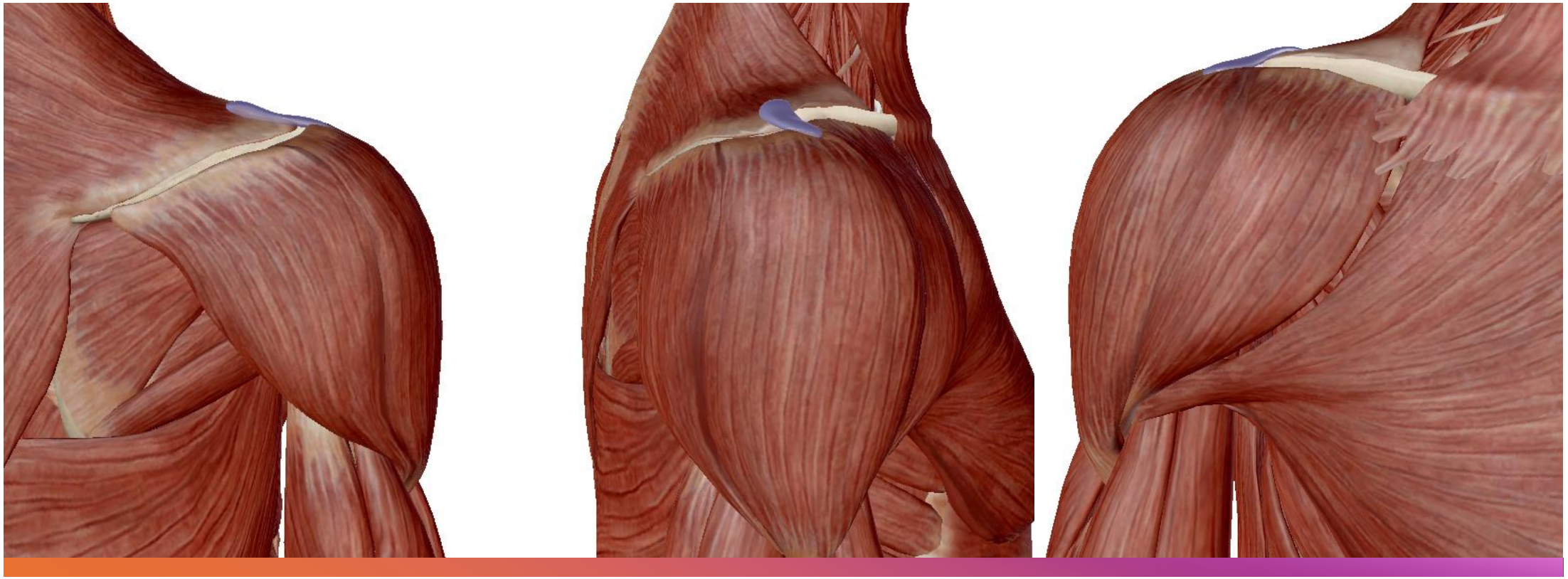


Muscles of the Pectoral Girdle and Arm

Muscles that position the pectoral girdle

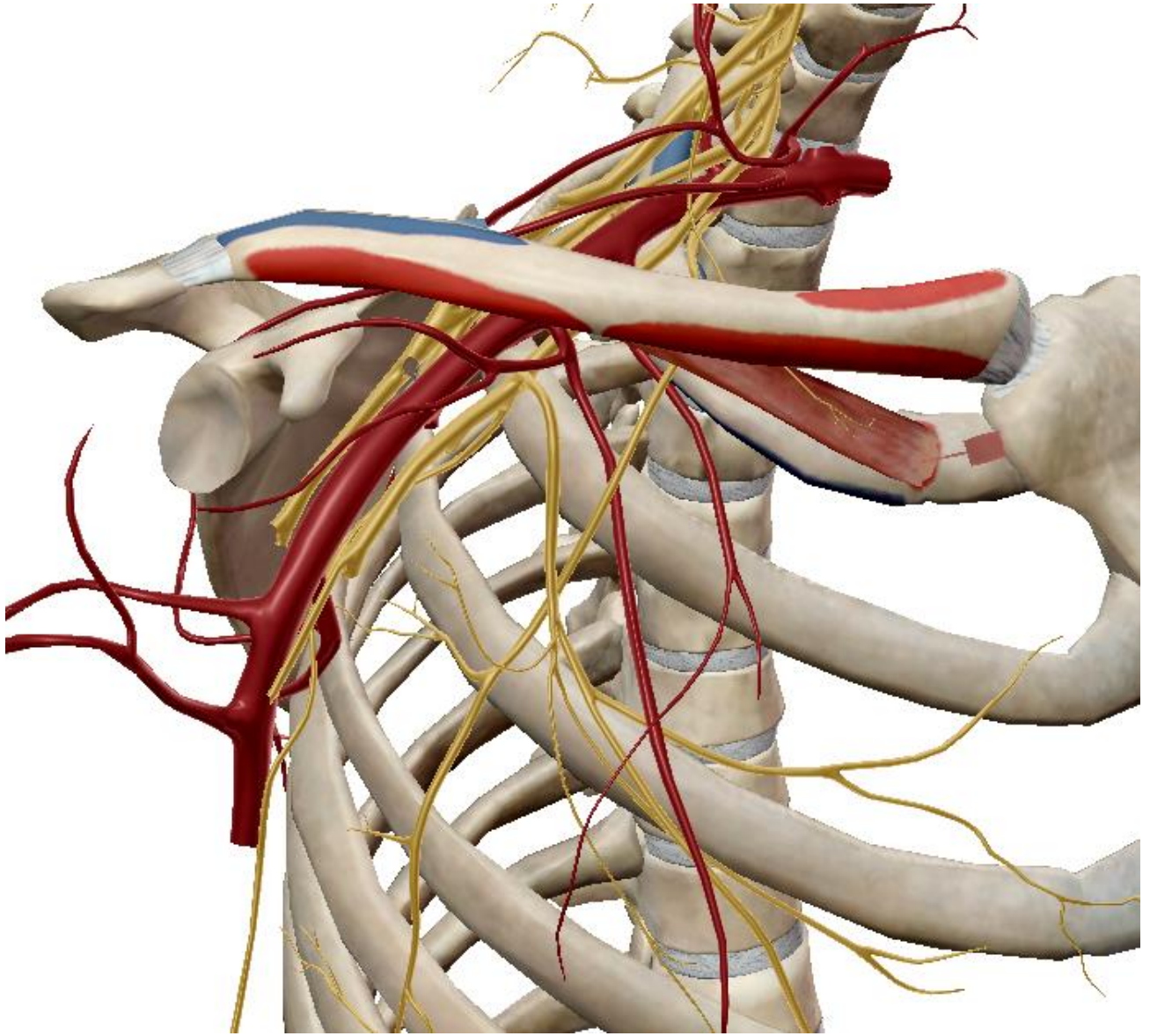
- **Anterior Thorax**
  - Pectoralis Minor
  - Serratus Anterior
  - Subclavius *(deep-seated)*
- **Posterior Thorax**
  - Trapezius
  - Rhomboid Minor
  - Rhomboid Major

- Toggle between the musculature icon and skeletal system icon on the app
- Identify the muscles that position the pectoral girdle.
- Select and identify individual muscles
- Explore their origins and insertions
- Discuss the actions of the muscles relative to their origins and insertions
- What nerves supply these muscles?
- Highlight each muscle and use the pin icon to explore the neurovascular supply!



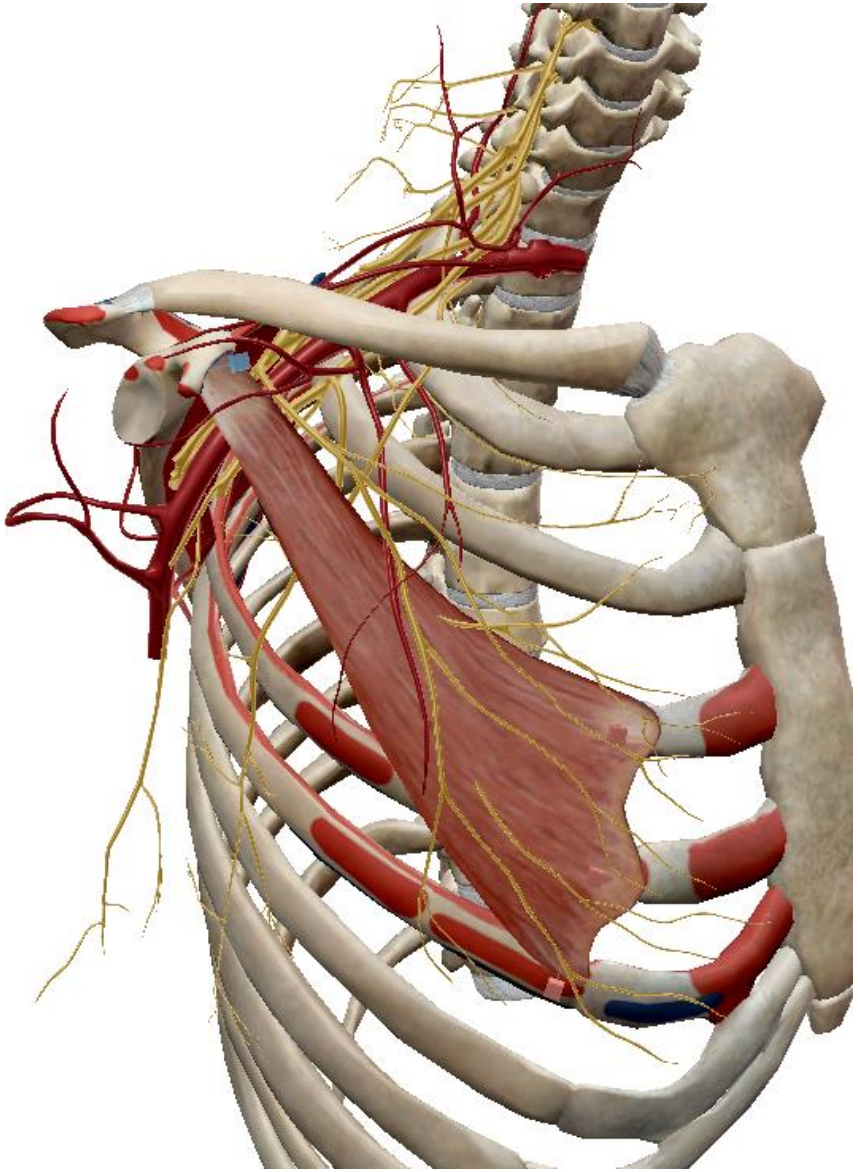
Explore the Posterior, Lateral, and Anterior Views

Subclavius



Pectoralis Minor

Explore muscle actions



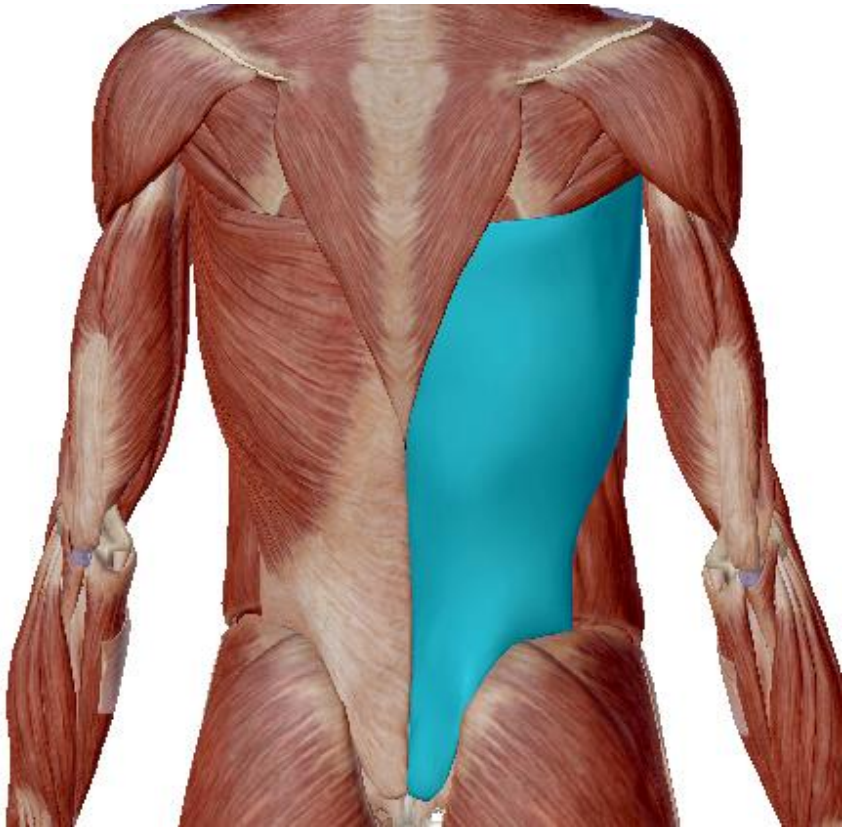
Muscles of the Pectoral Girdle and Arm

Muscles that move the Humerus

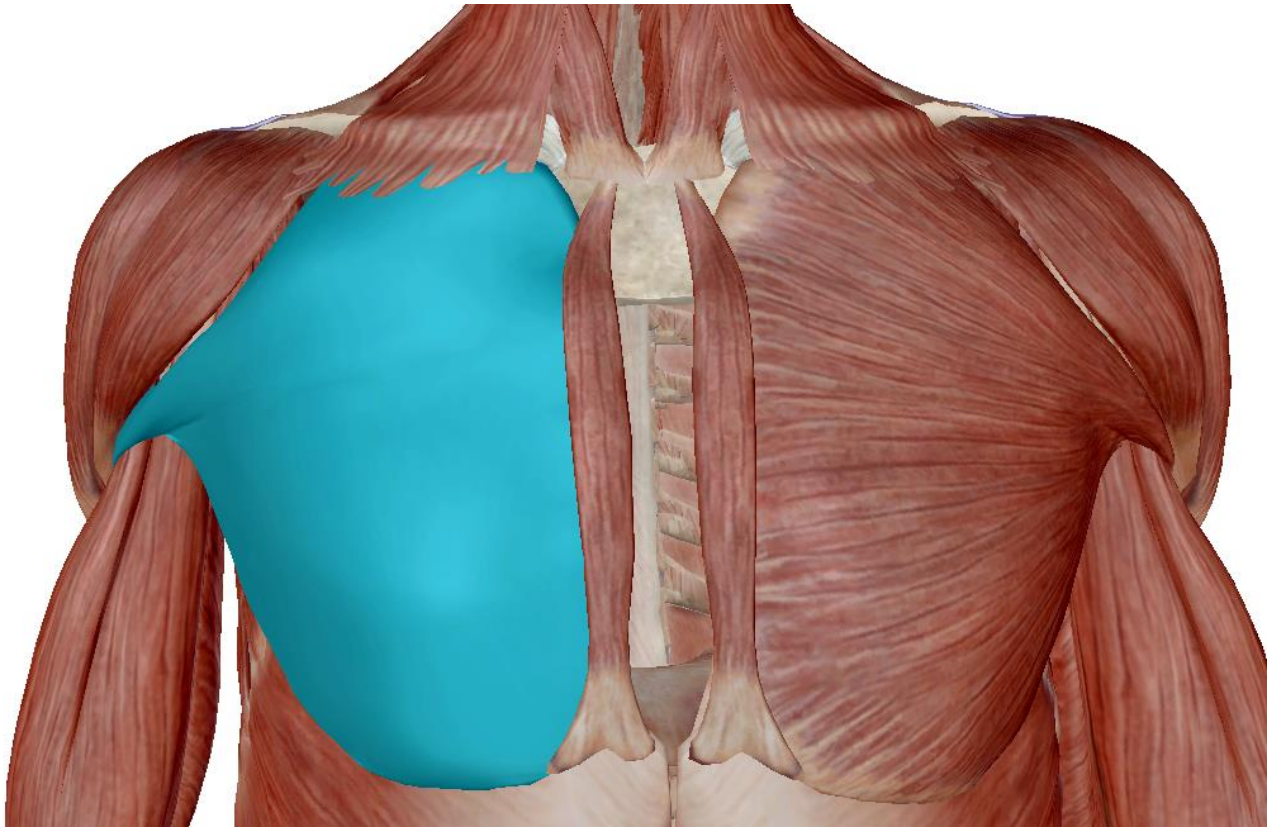
- **Axial Muscles**
  - Pectoralis Major
  - Latissimus Dorsi
- **Scapular Muscles**
  - Deltoid
  - Subscapularis
  - Supraspinatus
  - Teres Minor
  - Teres Major

- Toggle between the musculature icon and skeletal system icon on the app
- Identify the muscles that move the humerus.
- Select and identify individual muscles
- Explore their origins and insertions
- Discuss the actions of the muscles relative to their origins and insertions
- What nerves supply these muscles?
- Highlight each muscle and use the pin icon to explore the neurovascular supply!

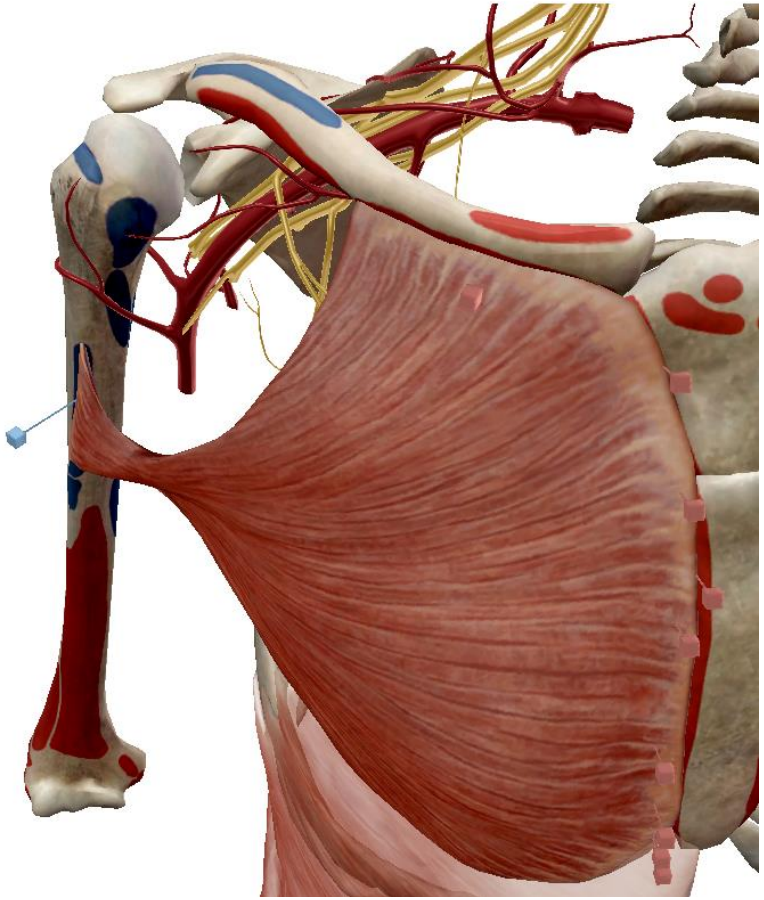
Latissimus Dorsi

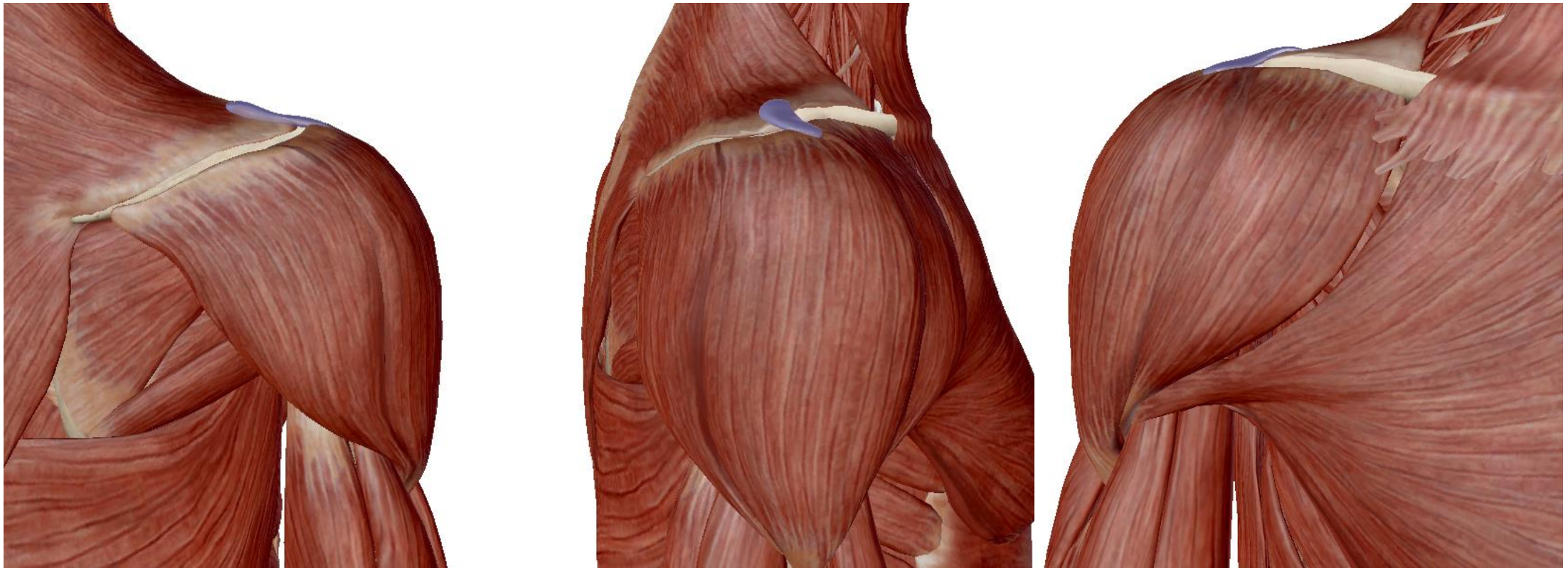


- Pectoralis Major



Pectoralis Major: Neurovascular supply and insertions





Explore the actions of the 3 heads of the Deltoid

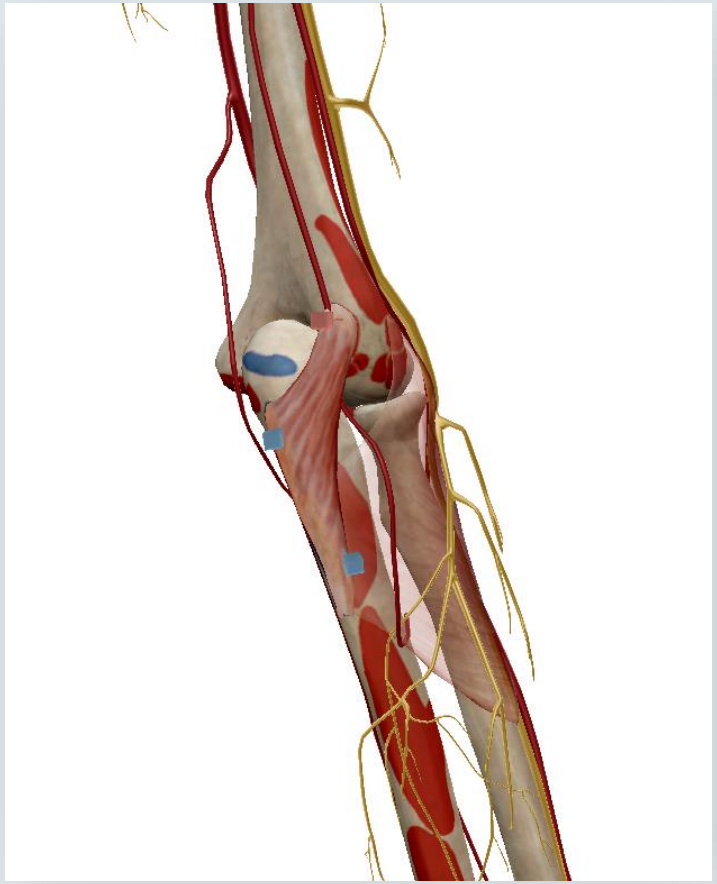
Muscles of the Pectoral Girdle and Arm

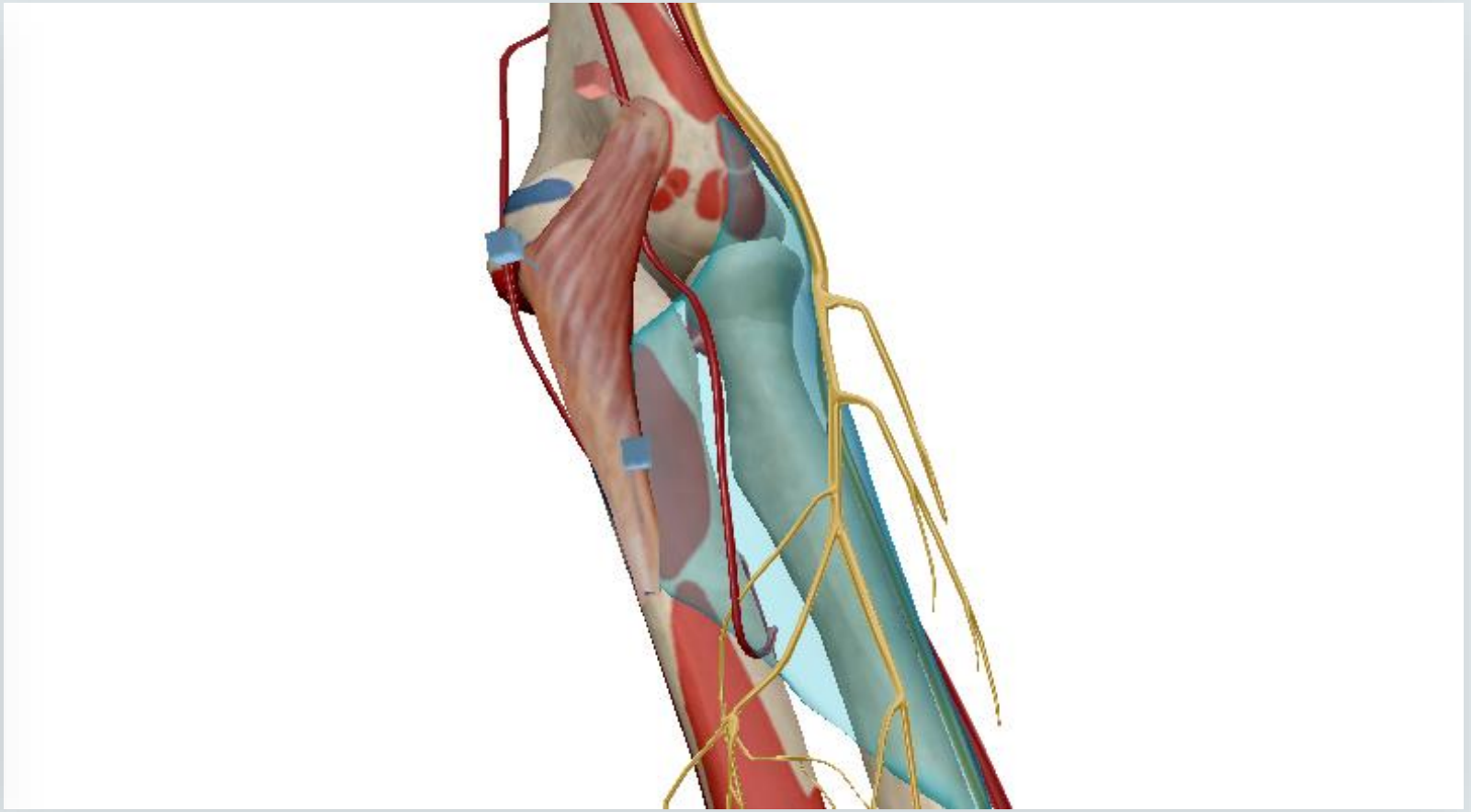
Muscles that move the Forearm

- Flexors
  - Bicep brachi
  - Brachialis
  - Brachioradialis
- Extensors
  - Triceps brachi
  - Anconeus
- Pronators
  - Pronator teres
  - Pronator quadratus *(Distal forearm)*
- Supinator
  - Supinator

- Toggle between the musculature icon and skeletal system icon on the app
- Identify the muscles that move the Forearm.
- Select and identify individual muscles
- Explore their origins, insertions and actions.
- Discuss the actions of the muscles relative to their origins and insertions
- What nerves supply these muscles?
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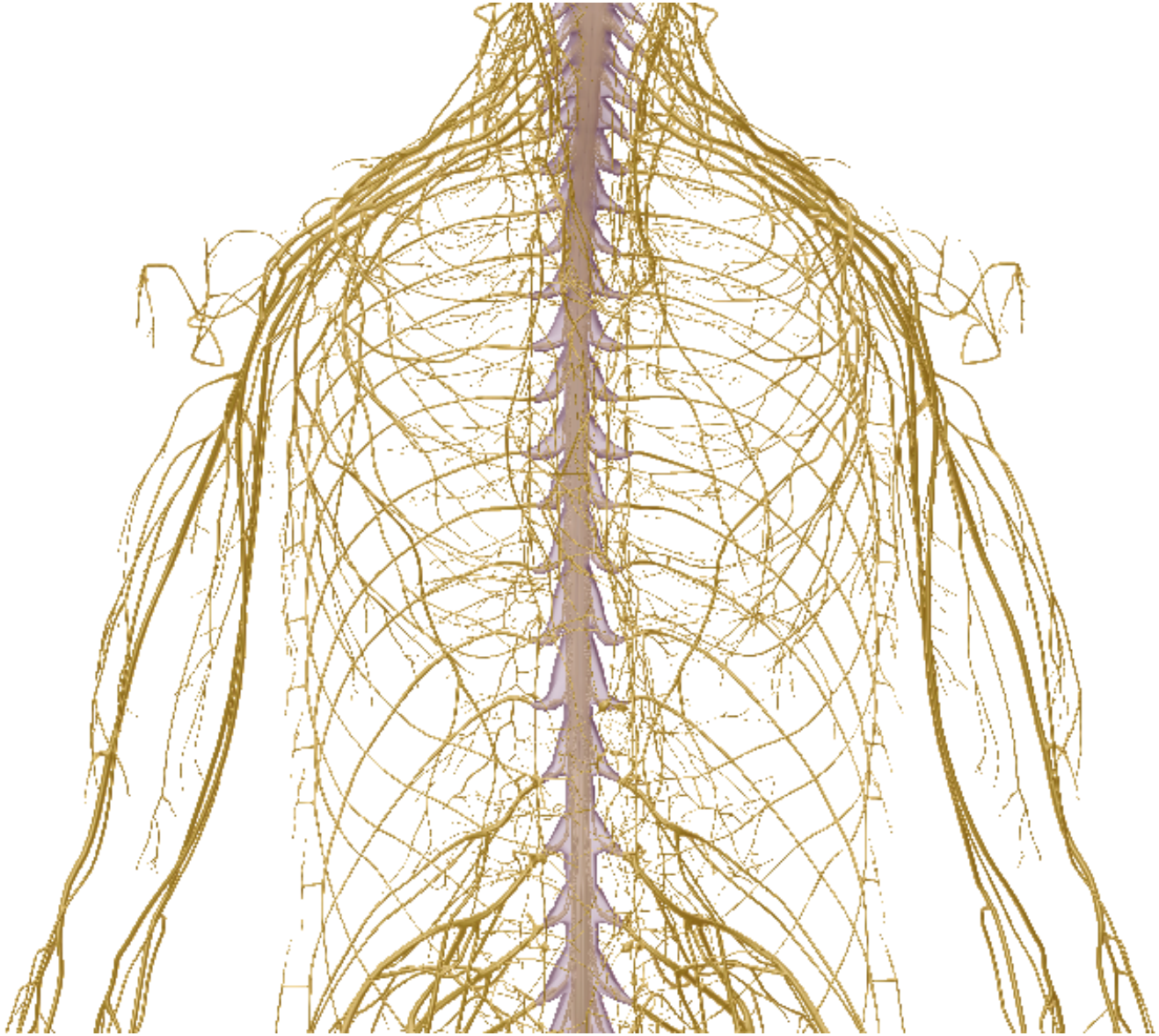
Anconeus and Supinator Muscles





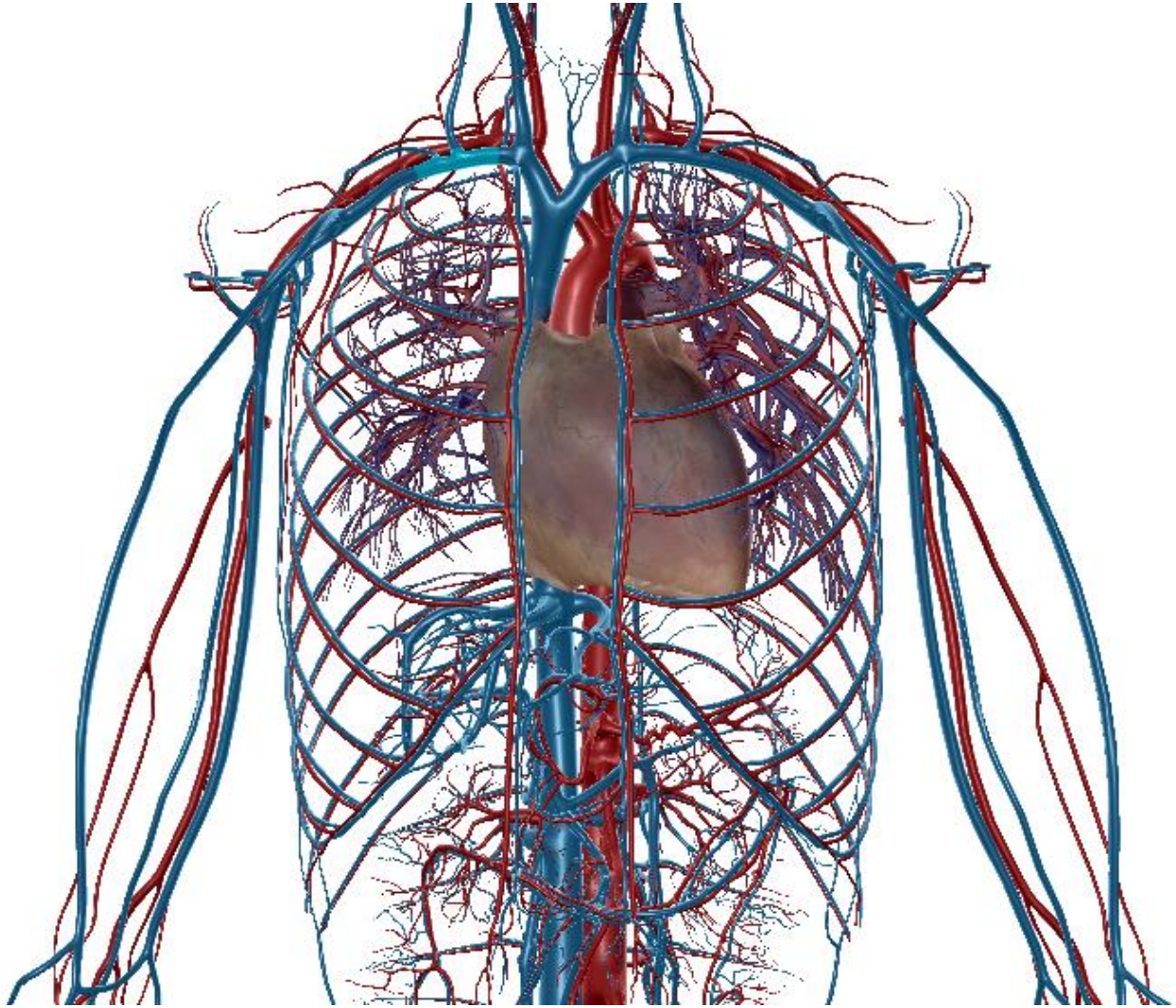
Neurovascular Structures of the Pectoral Girdle and Arm

- Select the **Nervous system** Icon (Brain)
- Deselect or Hide the skeletal system and musculature icons
- Identify the Nerve roots: C5 – T1
- Explore how they give rise to the Brachial Plexus:
  - Trunks
  - Divisions
  - Cords
  - Main Branches



Neurovascular Structures of the Pectoral Girdle and Arm

- Select the **Circulatory system** Icon (Heart)
- Deselect or Hide the Nervous system icon
- Identify the Blood Vessels: Arteries and Veins
- Explore how arteries originate from the Aorta to the Axillary artery and branch out.
- Explore how veins empty into the Axillary vein leading to the superior vena cava.





Questions